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National Center for Regenerative Soil & Sustainable Agriculture

Kingdom of Bahrein

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1. Executive Summary

This initiative provides for the establishment in the Kingdom of Bahrain of the first integrated scientific and industrial center in the GCC for the processing of sewage sludge (biosolids - wastewater treatment residues) into safe, fertile soil, with its subsequent use for agricultural crop cultivation.

The project establishes a closed-loop circular economy model in which wastewater treatment plant by-products are transformed into a strategic resource for the country's agricultural sector.

The Center is planned to be located within the BIW (Bahrain Investment Wharf) industrial park.

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2. Strategic Objectives

The initiative aligns with Bahrain Vision 2030, ESG (Environmental, Social, Governance) principles, and the Sustainable Development Goals (SDGs 6, 13, 15).

The project aims to achieve the following objectives:

→ Enhancement of Bahrain's food security and resilience
→ Introduction of innovative solutions in sustainable land management
→ Reduction of environmental impact from wastewater sludge
→ Establishment of a national scientific platform for soil regeneration
→ Development of an exportable model for GCC countries

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3. Project Structure

The project will be implemented on a 1.5-hectare land plot and includes three functional components:

A. Production Module

- Implementation of reagent-based sludge treatment technology
- Establishment of a fertile soil production center
- Compliance with GCC environmental and sanitary standards

B. Experimental Agricultural Site (1 ha)

- Formation of a 40 cm fertile soil layer
- Cultivation of agricultural crops
- Conducting agronomic and water-efficiency research
- Demonstration of the effectiveness of locally produced soil

C. Scientific and Laboratory Center (400-600 m2)

- Analytical research
- Scientific validation of results
- Training of specialists
- Establishment of a regional center of expertise

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Preliminary project map

- Experimental Agricultural Site
- Scientific and Laboratory Center
- Production Module

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4. Institutional Framework (Partnership Model)

Dr. Abdel Hadi
Scientific Supervisor of the Experimental Site

Arabian Gulf University (AGU)

- Scientific leadership of agronomic programs
- Conducting research and publications
- Academic expertise
- Integration of the project into AGU's scientific agenda

NIAD

- Strategic coordination
- Integration into national food security programs
- Interaction with relevant ministries

INOVEST / BIW

- Provision of land
- Access to wastewater treatment facilities
- Infrastructure support

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- Implementation of processing technology
- Technological supervision and support
- Ensuring environmental safety of the process

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5. Expected Outcomes

- Establishment of the first GCC soil regeneration center based on biosolids
- Creation of a local source of fertile soil
- Reduction of waste volumes sent to landfills
- Scientific validation of the effectiveness of the new model
- Strengthening Bahrain's position as a regional leader in sustainable agriculture

6. Strategic Significance

- The initiative transforms the wastewater sludge management system from a disposal model into a resource utilization model.
- The project integrates industrial infrastructure, innovative technology, and academic expertise into a unified national platform for sustainable development.

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